Partition coefficients for generalized peakon transfer matrices

Research proof and Lean formalization

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Abstract

We give all partition coefficients in the trace powers of a generalized peakon Lax matrix. A rank-two determinant calculation reduces its characteristic determinant to one Hamiltonian generating polynomial. The formal trace-log identity then gives a closed factorial formula, including the two coefficient families proposed by Avan, Frappat and Ragoucy. The proof applies to arbitrary coherent complex particle amplitudes and, literally, to the square-root matrices for nonnegative real momenta.

Scope and review status. The mathematical and formal statements below use the explicitly stated amplitude domains. Independent preliminary source and mathematical review confirms the coherent square-root convention used below. Independent final source, mathematical, formal, manuscript and bounded novelty review has passed. No worldwide priority claim is made.

Let $c_i \in \mathbb{C}$, $q_i \in \mathbb{R}$ $(1 \le i \le N)$, and $\nu, \gamma, z \in \mathbb{C}$, with $2z + \gamma \neq 0$. Define the actual hyperbolic matrices

$$A_{ij} = c_i c_j \sinh\Big(\frac{\nu}{2}(q_i - q_j)\Big), \qquad T_{ij} = c_i c_j \cosh\Big(\frac{\nu}{2}(q_i - q_j)\Big), \qquad S_{ij} = \operatorname{sgn}(q_i - q_j) A_{ij},$$
$$\Lambda = \frac{z^2 - 1}{2z + \gamma}, \qquad \xi = \frac{z^2 + \gamma z + 1}{2z + \gamma}, \qquad L = S + \Lambda T + \xi A.$$

Here $\operatorname{sgn}(0) = 0$. Put $\tau_m = \operatorname{tr}(L^m)$ and, exactly as in [1, eq. (5.17)],

$$h^{(k)} = k \operatorname{tr}\Big((T + \tfrac{\gamma}{2}A)(S + A)^{k-1}\Big) \qquad (k \ge 1). \tag{1}$$

Theorem 1. *For every $m \ge 1$,*

$$\tau_m = \sum_{j=1}^{m} \Lambda^j \sum_{\substack{k_1 \ge \cdots \ge k_j \ge 1 \\ k_1 + \cdots + k_j = m}} \frac{m(j-1)!}{(\prod_{i=1}^{j} k_i)(\prod_{s \ge 1} r_s!)} \prod_{i=1}^{j} h^{(k_i)}, \tag{2}$$

where $r_s = \#\{i : k_i = s\}$. In particular, the proposed coefficient families are

$$a_1(m) = 1, \qquad a_m(1, \dots, 1) = 1, \qquad a_{j+1}(m - j, \underbrace{1, \dots, 1}_{j}) = \frac{m}{m - j} \quad (0 \le j < m - 1). \tag{3}$$

No independence or uniqueness of these monomials for fixed N is asserted; the displayed universal coefficients give a valid expansion for every N.

Corollary 1. *For arbitrary $p_i \ge 0$, Theorem 1 holds for the literal factors $\sqrt{p_i p_j}$ in place of $c_i c_j$, with the same $h^{(k)}$ and all real particle positions, including ties.*

Indeed choose $c_i = \sqrt{p_i}$; then $\sqrt{p_i p_j} = \sqrt{p_i}\sqrt{p_j}$.

The determinant reduction

First reorder the positions so that $q_1 \geq \cdots \geq q_N$, and set

$$u_i = c_i e^{\nu q_i/2}, \qquad v_i = c_i e^{-\nu q_i/2}, \qquad B = S + A.$$

Expanding sinh and cosh gives

$$A = \tfrac{1}{2}(uv^t - vu^t), \qquad T = \tfrac{1}{2}(uv^t + vu^t), \qquad B_{ij} = \begin{cases} u_i v_j - v_i u_j, & i < j, \\ 0, & i \geq j. \end{cases}$$

These identities remain valid when positions tie, since then $A_{ij} = 0$. Thus $B^N = 0$, $\det(I - tB) = 1$, and

$$R = (I - tB)^{-1} = \sum_{k=0}^{N-1} t^k B^k, \qquad B - B^t = uv^t - vu^t.$$

All equalities are polynomial identities over a characteristic-zero field. Write

$$a = \frac{\Lambda + \xi - 1}{2}, \quad b = \frac{\Lambda - \xi + 1}{2}, \quad L = B + auv^t + bvu^t,$$

and put $P = v^t R u$, $Q = u^t R v$, $C = PQ - (v^t R v)(u^t R u)$. The classical rank-two determinant lemma gives

$$\det(I - tL) = 1 - t(aP + bQ) + t^2 abC. \tag{4}$$

The resolvent identity

$$R - R^t = tR(B - B^t)R^t$$

gives $P - Q = tC$ after multiplication by v^t and u. Also

$$\xi^2 = \Lambda^2 + \gamma\Lambda + 1, \qquad a + b = \Lambda, \qquad ab = \frac{\xi - 1}{2} - \frac{\gamma\Lambda}{4}.$$

Substitution in (4) cancels the $\xi - 1$ terms and proves

$$\det(I - tL) = 1 - \Lambda H(t), \qquad H(t) = t \operatorname{tr}\left((T + \tfrac{\gamma}{2}A)R\right) = \sum_{k=1}^{N} \frac{h^{(k)}}{k} t^k. \tag{5}$$

This is the source-specific reduction. All determinant, trace and Hamiltonian expressions are invariant under simultaneous permutation of the particle indices, so (5) holds without an ordering assumption. Nilpotence also proves $h^{(k)} = 0$ for $k > N$.

Every trace order and every partition

The classical formal trace-log identity, with constant determinant coefficient one, is

$$-\log\det(I - tL) = \sum_{m \geq 1} \frac{\operatorname{tr}(L^m)}{m} t^m.$$

For clarity, it can be proved by constructing the matrix geometric series $(I - tL)^{-1}$ and applying Jacobi's determinant derivative identity. Both sides then have the same derivative and zero constant term. Combining this identity with (5), using $H(0) = 0$, yields

$$\sum_{m \geq 1} \frac{\tau_m}{m} t^m = \sum_{j \geq 1} \frac{\Lambda^j}{j} H(t)^j. \tag{6}$$

Only $j \leq m$ contributes to the coefficient of t^m. A fixed partition with multiplicities r_s has $j!/\prod_s r_s!$ orderings in $H(t)^j$. Its contribution is therefore

$$\frac{m}{j} \frac{j!}{\prod_s r_s!} \prod_{i=1}^{j} \frac{h^{(k_i)}}{k_i},$$

which proves (2). The three evaluations in (3) follow by direct cancellation. The restriction $j < m-1$ ensures $m - j > 1$; the all-ones case has its own multiplicity factor.

Source relation and formal scope

The Lax matrices originate in [3, 2]. Avan–Frappat–Ragoucy [1, Conjecture 5.2, eqs. (5.20)–(5.21)] propose the partition expansion and the families (3), leaving the other coefficients undetermined. Theorem 1 supplies all coefficients for the coherent-amplitude family; Corollary 1 uses their displayed square-root factors literally for $p_i \geq 0$. The source's notation $\sqrt{p_i p_j}$ uses coherent per-particle roots: its identity $T_{ii} = p_i$, from eqs. (2.2) and (5.6), and its degree-m homogeneity require this convention. Thus $c_i^2 = p_i$ recovers its family, including complex coherent roots for negative momenta. Arbitrary c_i in Theorem 1 covers this convention; the corollary additionally proves the nonnegative principal-root reading directly. The rank-two determinant lemma, Jacobi identity, formal trace-log identity and multinomial theorem are classical ingredients, not separate discoveries.

The Lean development has 121 declarations in 19 modules, using Lean 4.34.1 and mathlib revision
`d13f23b723b8a846827a245b89c10fc7d3f11612`. It proves the determinant collapse, the formal trace-log and multinomial bridges, all-order support truncation, the actual complex hyperbolic matrix identities, permutation sorting, and both coefficient families. The closed source endpoint and its literal nonnegative specialization are:

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source_conjecture_372008
nonnegative_source_conjecture_372008
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The isolated strict replay passes with only `propext`, `Classical.choice` and `Quot.sound`. Independent final review passes all five gates. Discovery acceptance additionally requires a fresh replay from the extracted proof packet; no global priority or optimality claim is made.

References

[1] J. Avan, L. Frappat and E. Ragoucy, *Quadratic Poisson brackets for the Camassa–Holm peakons*, J. Phys. A: Math. Theor. **59** (2026), 245202. doi:10.1088/1751-8121/ae7683; arXiv:2512.11066v3.

[2] J. Avan, L. Frappat and E. Ragoucy, *A new integrable structure associated to the Camassa–Holm peakons*, SciPost Phys. **15** (2023), 228. doi:10.21468/SciPostPhys.15.6.228.

[3] O. Ragnisco and M. Bruschi, *Peakons, R-matrix and Toda lattice*, arXiv:solv-int/9509012v1 (1995).